THEORY ASSIGNMENT – 3

Frontend Development • JavaScript

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1. Difference between frontend and backend JavaScript

Frontend JavaScript runs in the browser and makes web pages interactive. Backend JavaScript runs on a server and handles requests, databases, and application logic.

2. What is DOM?

DOM (Document Object Model) represents an HTML page as a tree of elements. JavaScript uses it to read or change page content, attributes, and styles.

3. What is an event in JavaScript?

An event is an action such as a click, key press, or form submission that JavaScript can detect and respond to.

4. What is the event loop?

The event loop helps JavaScript handle queued callbacks. After the current synchronous code finishes, it allows pending asynchronous callbacks to run when the call stack is clear.

5. What is hoisting?

Hoisting is JavaScript's processing of declarations before execution. Function declarations can be called earlier; `var` is initialized as `undefined`, while `let` and `const` cannot be accessed before declaration.

6. What is event bubbling?

Event bubbling is when an event starts on a nested element and then propagates to its parent elements. `event.stopPropagation()` can stop it.

7. Named, anonymous, and arrow functions

A named function has a name (e.g. `function greet() {}`). An anonymous function has no function name (e.g. `function() {}`). An arrow function uses `=>` (e.g. `() => {}`). Arrow functions inherit `this` from their surrounding scope.

8. Explain `for...of` and `for...in`

`for...of` loops through values of iterables such as arrays and strings. `for...in` loops through enumerable property keys, commonly object keys.

9. Explain any five array methods

push() adds an item at the end. **pop()** removes the last item. **shift()** removes the first item. **unshift()** adds an item at the beginning. **map()** creates a new array by transforming each item.

10. What is a callback function?

A callback is a function passed to another function to be called later. Example: `button.addEventListener("click", function() { alert("Clicked"); });`